

VBT-Based Injury Prevention & Recovery Coach

SIH 2026 — SIH26213 | AICTE | Fitness & Sports | Hardware Category

1. Expert Rating (Honest Assessment)

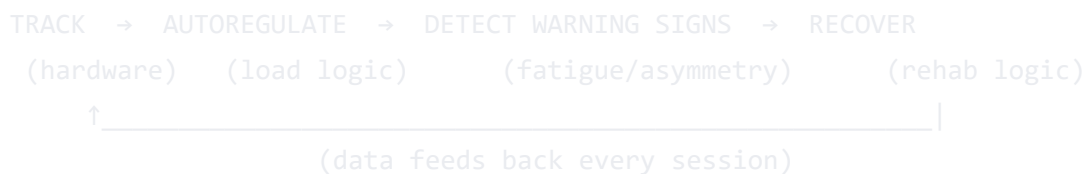
Criteria	Rating	Notes
Problem relevance	9/10	Injury is the #1 reason athletes/lifters stall or quit. Real, validated pain point.
Novelty in India context	8/10	VBT devices exist globally (GymAware, Output Sports, Vmax Pro) but cost ₹1.5-4L+. An affordable, India-built version is a strong "why now, why us" story.
Technical feasibility (team-dependent)	7/10	Core loop (IMU → velocity → load logic) is very achievable. CV and rehab logic are feasible <i>if scoped correctly</i> — see roadmap.
Demoability	9/10 (if scoped right)	Live velocity tracking + load recommendation is verifiable in the room — the strongest kind of SIH demo.
Defensibility in Q&A	8/10	Strong <i>if</i> we avoid vague AI claims and lean on the S&C-coach-built rehab logic + real sports science (load-velocity profiling).
Risk of overreach	High if all features are promised as "built"	Must clearly separate what's built vs. roadmap.

Overall: This is a strong, winnable SIH idea — but only if we present it as a focused product with a clear phased roadmap, not a feature list. Judges reward technical depth in one thing over shallow coverage of four things.

2. The Core Story (memorize this)

"A wearable that helps you train smart enough to avoid getting hurt — and coaches you back when you do."

We are not building "an AI fitness app." We are building a **closed-loop training safety system**:



Every feature we build must plug into this loop. If a feature doesn't clearly belong in one of these four stages, it doesn't go in this version of the product.

3. Why This Matters (the problem)

- Commercial VBT devices (GymAware ~₹3L+, Output Sports, Vmax Pro) are priced for elite pro teams — completely inaccessible to Indian college athletes, small academies, and individual lifters/coaches.
 - Athletes and even coaches often select training loads by feel or by generic percentage-based programs, not by real-time readiness — a major contributor to overtraining injuries.
 - When injuries do happen, most athletes get generic rehab advice (YouTube, forums) with no connection to their actual training history — recovery becomes guesswork, and re-injury rates stay high.
 - There is no affordable, India-built product that connects **training data** → **injury risk** → **recovery** as one continuous system.
-

4. What We Are Building — Feature Breakdown

4.1 Hardware: The Tracker (Stage 1 — TRACK)

- Compact, magnetic-mount device attached to a barbell (or worn for jump tests)
- **MCU:** Sseed XIAO nRF52 (BLE-capable, low power)
- **Sensor:** BMI270 IMU (SPI) — accelerometer + gyroscope
- Battery + single button for start/stop of a set
- Streams raw motion data over BLE to the companion app
- **Output metrics:** mean concentric velocity, peak velocity, bar path (later phase), rep count, rest time between sets

Known hard problem to solve: IMU double-integration drift. We will use **Zero-Velocity Update (ZUPT)** correction — detecting the static moment at the top/bottom of each rep to reset velocity error — combined with a complementary or Kalman filter fusing accelerometer + gyro data. This is a real engineering challenge and we should be ready to explain it clearly in Q&A — it shows judges we understand the hard part of the problem, not just the pitch.

4.2 Load Selection Engine (Stage 2 — AUTOREGULATE)

This is our **strongest, most demo-ready pillar**. Based on established sports science (load-velocity profiling), not experimental AI:

- Athlete performs a few warm-up reps at increasing submaximal loads
- App plots **velocity vs. load** and fits a linear regression
- From this profile, the app estimates **e1RM (estimated 1-rep max)** — *without the athlete ever needing to attempt an actual max lift*, which is itself one of the highest-injury-risk moments in training. Removing the need for max-outs is a genuine injury-prevention win we can say out loud.
- App recommends load for a target **RPE (Rate of Perceived Exertion)** or velocity zone (e.g., "for RPE 8 strength work today, use ~82.5kg")
- Within a session, **velocity loss %** across sets flags fatigue in real time (e.g., "velocity dropped 22% from set 1 — consider reducing load or stopping the set") — this is a well-documented, validated fatigue-management method used by real strength coaches, not a black box

4.3 Movement Asymmetry Detection (Stage 3 — DETECT WARNING SIGNS, Phase 2)

Reframed from "form analysis" / "gait analysis" — those are the wrong terms for what we're building.

- Phone-camera pose estimation (MediaPipe/OpenPose) tracking key points during a lift
- Detects **bar path drift, left-right timing asymmetry, depth consistency** — documented pre-injury compensation patterns
- **Cross-validates against hardware velocity data**: if both the camera and the IMU show a velocity/asymmetry issue on the same rep, that's a much stronger signal than either source alone — this is a nice technical story (sensor fusion for confidence)
- This is a **Phase 2 feature** — described and mocked up in the pitch, not fully built for internal round or even initial MVP unless time allows

4.4 Rehab Assistant (Stage 4 — RECOVER)

- **NOT** framed as "self-trained AI." Framed accurately as: *a decision-support system designed by an S&C coach with 3 years of hands-on injury rehab experience, encoding rehab-science protocols, personalized using the athlete's own tracked velocity/asymmetry/load history.*
- Built as a **rule-based / decision-tree expert system**, not machine learning — explainable, defensible, and genuinely useful
- Example logic: *"Velocity asymmetry >15% between limbs, sustained over 3 sessions, combined with reported pain location X → flag pattern Y → recommend rehab protocol Z."*

- Takes input from: previously tracked load/velocity history + a structured intake questionnaire (pain location, onset, aggravating movements)
- Outputs a rehab plan grounded in the athlete's actual data — not a generic chatbot answer
- **Safety framing is mandatory:** this is a *coach-assist* tool, not a medical diagnosis. Every rehab output should carry a clear disclaimer and encourage professional consultation for anything beyond minor/common issues. This protects the team and is what real products do.

4.5 Dashboard / App (glue layer)

- Session history, velocity graphs per exercise, fatigue trend lines, e1RM tracking over time
- Coach view vs. athlete view (dual-user design — coaches manage multiple athletes' data)

5. Who This Is For

- **College/academy strength & conditioning coaches** managing multiple athletes without expensive lab equipment
- **Individual serious lifters/athletes** who want data-driven training without elite-level budgets
- **Physiotherapists/rehab professionals** who want objective tracked data instead of relying only on patient self-report

6. Competitive Landscape (know this cold for Q&A)

Product	What they do	Gap we fill
GymAware / Output Sports	Pro-level VBT hardware	Very expensive, not accessible in India, no rehab loop
Vmax Pro / gPath	Compact VBT trackers	Load tracking only — no injury-prevention or rehab intelligence layer
Stance Fitness / Lift.app.ai	Camera-based rep counting/form	No hardware-grade velocity data, no rehab connection
Us	Affordable IMU hardware + load autoregulation + camera cross-validation + coach-built rehab logic, all in one closed loop	—

7. Phased Roadmap (say this explicitly in the pitch)

Phase	What's included	Status
Phase 0 — Internal round (Sept 21)	Working hardware prototype: IMU → BLE → velocity readout + graph	Build now
Phase 1 — Main SIH PPT submission	Load selection engine (e1RM estimation, velocity-loss fatigue flag), basic app UI	Design + partial build
Phase 2 — Post-selection MVP	Full app, session history, coach/athlete dashboards, rehab decision-support system (rule-based)	Build if selected
Phase 3 — Future/vision	Camera-based movement asymmetry detection, sensor fusion with hardware data	Roadmap slide only — do not claim as built

Golden rule for every teammate: never present a feature as "working" unless it is actually working on the day of presentation. Roadmap slides are fine and expected — false demos are not.

8. Key Risks & How We're Handling Them

1. **IMU drift** → ZUPT + sensor fusion (see 4.1). Own this technical challenge openly.
2. **Overclaiming "AI"** → Be precise: load engine = statistical regression (real science), rehab tool = rule-based expert system built by our domain expert, not ML. This is *more* credible to judges, not less.
3. **Scope creep** → Strict phase separation (Section 7). If it's not in the current phase, it doesn't get built yet, no matter how tempting.
4. **Liability on rehab advice** → Explicit "coach-assist, not medical diagnosis" disclaimer, always visible in the rehab module.
5. **Team bandwidth** → Confirm who owns firmware/BLE, who owns app/UI, who owns the load-velocity math, who owns rehab logic content (coach's domain knowledge). No feature should be "everyone's responsibility."

9. Elevator Pitch (30 seconds, memorize)

"Elite athletes use velocity-based training devices costing lakhs of rupees to train smart and avoid injury. Everyone else trains blind. We built an affordable wearable that tracks bar speed in real time, tells you exactly what load to lift and when to back off before you get hurt — and if you do get injured, gives you a recovery plan built on your own training data by a real strength coach's expertise. One device, one loop: track, train safe, recover smart."